

WASP-52b

R-0329

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_221108 · created 2026-07-18T04:26:15+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459891.6682 +/- 0.0072 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.063
Transit depth (Rp/Rs)^2	3.92 +/- 2.49 [%]
Orbital Inclination (inc)	80.94 +/- 2.76
Airmass coefficient 1 (a1)	0.054 +/- 0.033
Airmass coefficient 2 (a2)	2.5 +/- 1.28
Scatter in the residuals of the lightcurve fit is	55.67 %
Best Comparison Star	None
Optimal Aperture	2.85
Optimal Annulus	9.04
Transit Duration (day)	0.028 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.198 ± 0.063 vs 0.1646 (deviation 0.42σ)
Duration fit vs published	0.028 d vs 0.0758 d (deviation 1.21σ)
Mid-time O–C	-1.2 min
Depth SNR	2.5
Residual scatter	55.67 %

Light curve

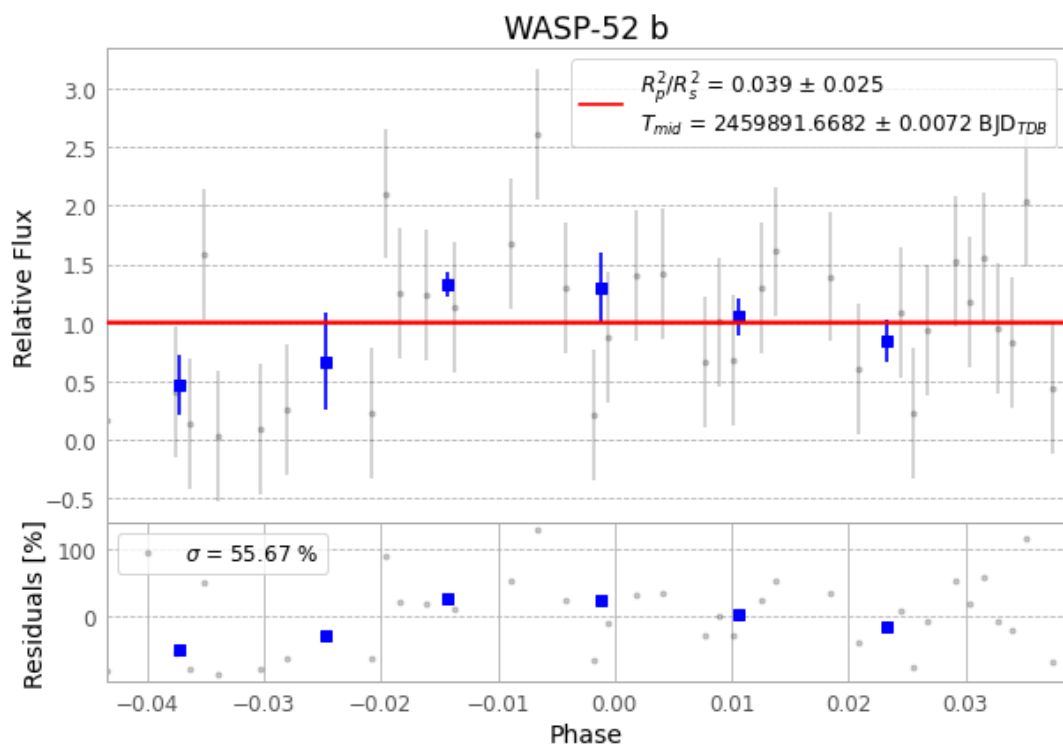


Figure 1 — FinalLightCurve_WASP-52 b_2022-11-07.png (SHA-256 8fe0e2900d18035e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [295, 59], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2d0ac648664f5b12...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a3bbcf8

Data provenance

76 FITS frames (50 MB), WASP-52221108020615.FITS → WASP-52221108055115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-221108.log (6a1db6c64f54...), FinalLightCurve_WASP-52 b_2022-11-07.png (8fe0e2900d18...), FinalLightCurve_WASP-52 b_2022-11-07.csv (93cfb8131db2...)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 04:26Z.